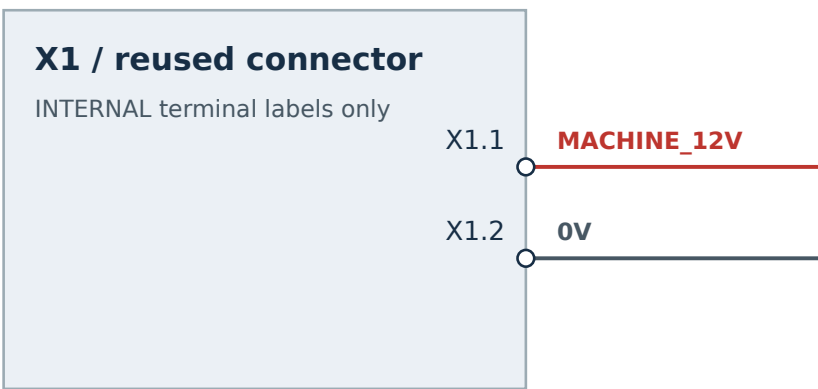
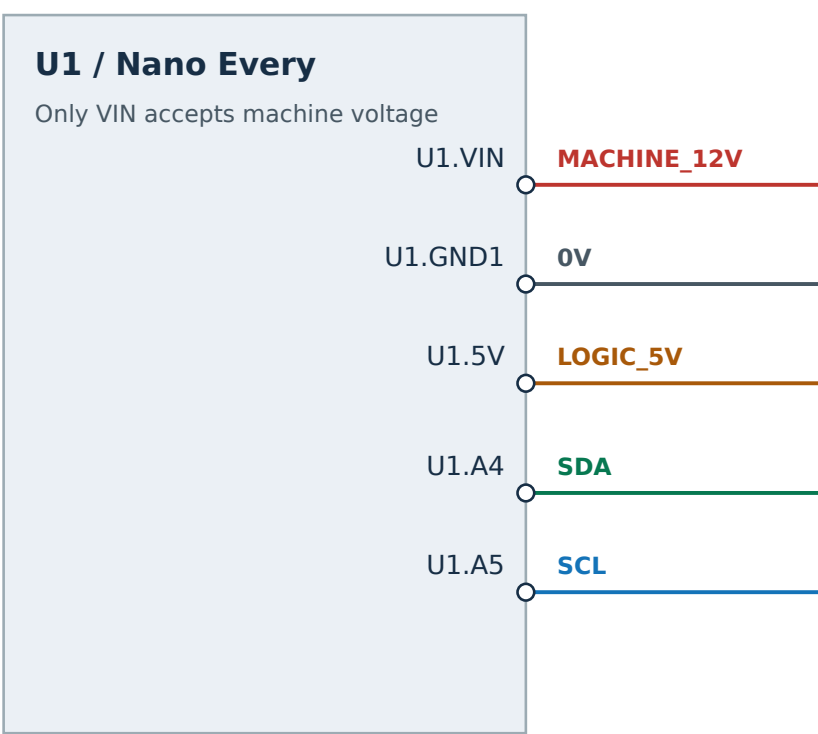


01 / POWER AND LOCAL BUS

Supply domains, module grounds, I2C addresses and Nano connections



X1.3 = G; X1.4 = H; X1.5 = J.
Actual machine cavities: H1 HOLD.
No additional fuse block: H2 HOLD.



Local HSD bypass at each VIN/GND: 47 uF / 35 V (+ to VIN) in parallel with 100 nF V.
USB can energize the logic rail. Disconnect machine, valve and water before USB servicing.

PARALLEL SUPPLY CONNECTIONS

MACHINE_12V -> TB12 -> U1 VIN + HSD1/2/3 VIN
0V -> TB0 -> all power/signal returns
LOGIC_5V -> TB5 -> U2 VCC, U3 VCC, HSD V,
O1/O2 HV, FM1 RED
HSD load GND also grounds its logic internally.
Do not carry lamp current through a logic header.
U2 OUT_GND = U2 GND internally.
U1 GND2 = U1 GND1 internally.
U4 relay 5 V must NOT connect to LOGIC_5V.

LOCAL I2C - 100 kHz

SDA: U1 A4 -> U2 SDA + HSD1/2/3 D
SCL: U1 A5 -> U2 SCL + HSD1/2/3 C
U2: 0x58; HSD pairs: 0x60/61, 0x62/63, 0x64/65
Use U2 onboard 4.7k pull-ups only.
Open all HSD SDA/SCL pull-up jumpers.
All bus wiring remains inside enclosure.

02 / G, H AND J INPUTS

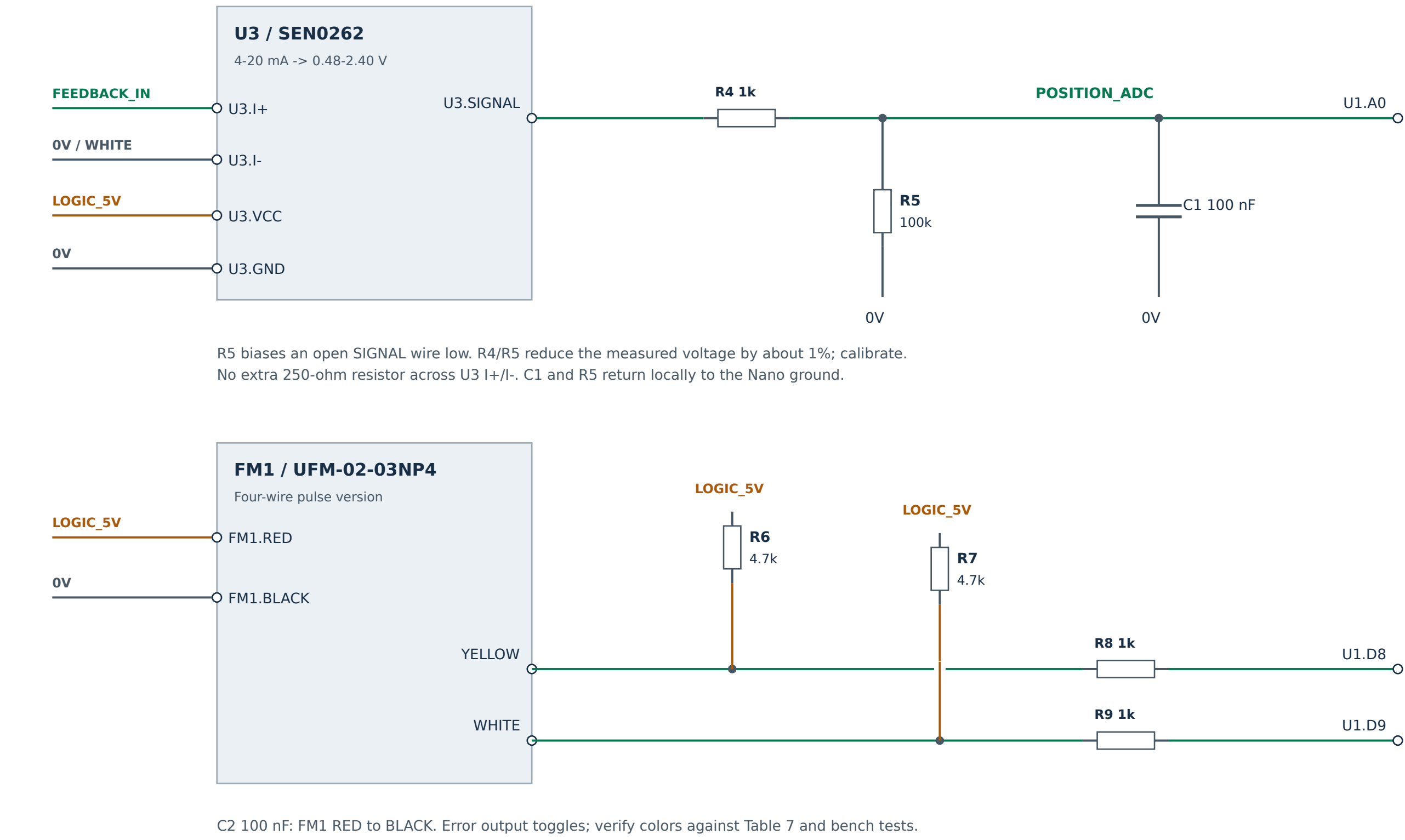
SparkFun BOB-09118 v1.2: HV terminal receives 5 V, not machine voltage



O1/O2 HV -> LOGIC_5V; HV-GND and INPUT_GND -> 0V. O2.IN2 -> INPUT_GND.
O2.OUT2 and both JP2 pin 1 pads unused. HIGH machine input -> HIGH Arduino input.
The three diodes are across the input LED circuits, after the external series resistors.

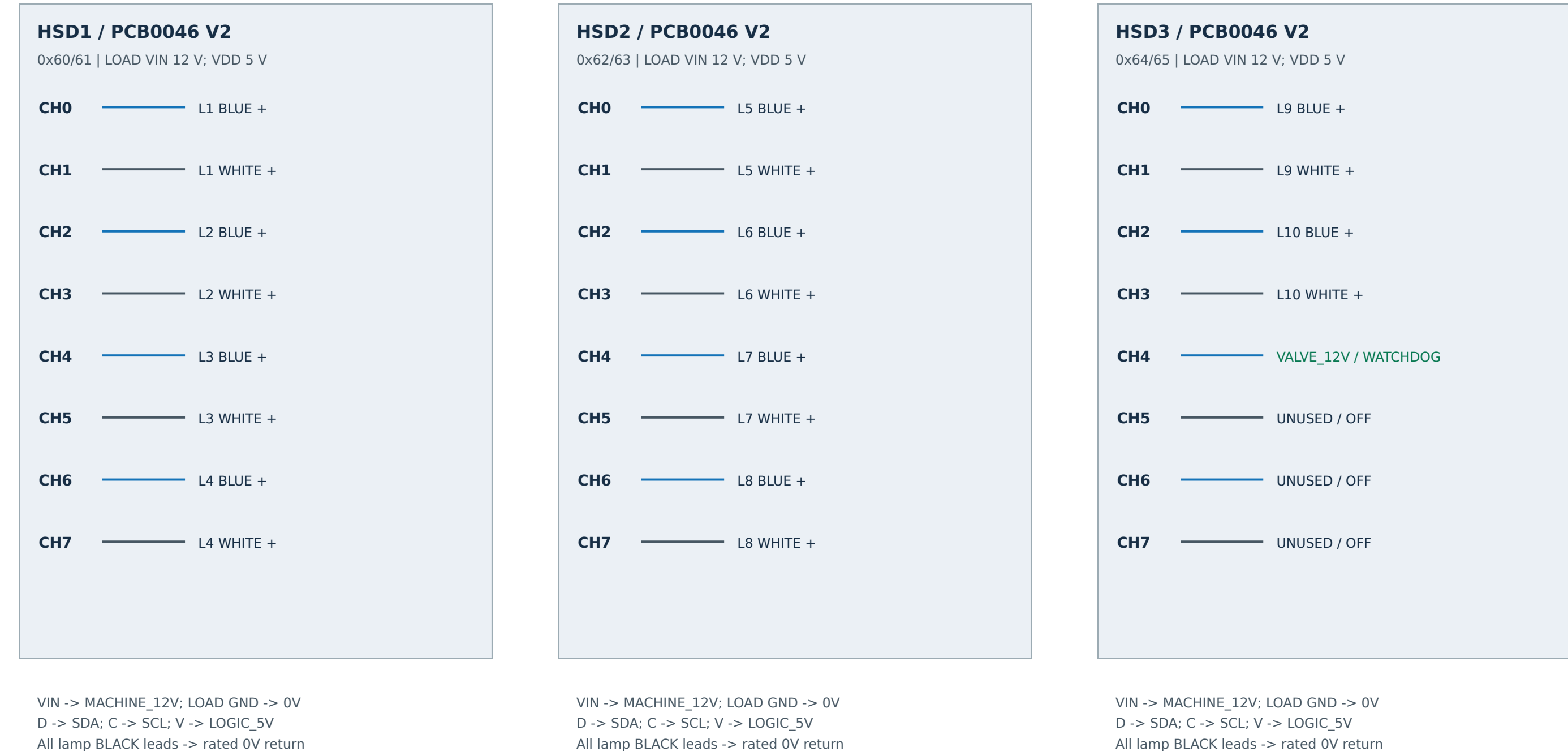
04 / ANALOG POSITION AND FLOW METER

No 12 V connection anywhere on this sheet; U2 current-loop output is not an ADC signal



05 / LAMP OUTPUTS AND BOARD CONFIGURATION

Use the printed channel numbers. Each blue/white lamp needs TWO separate switched positives.



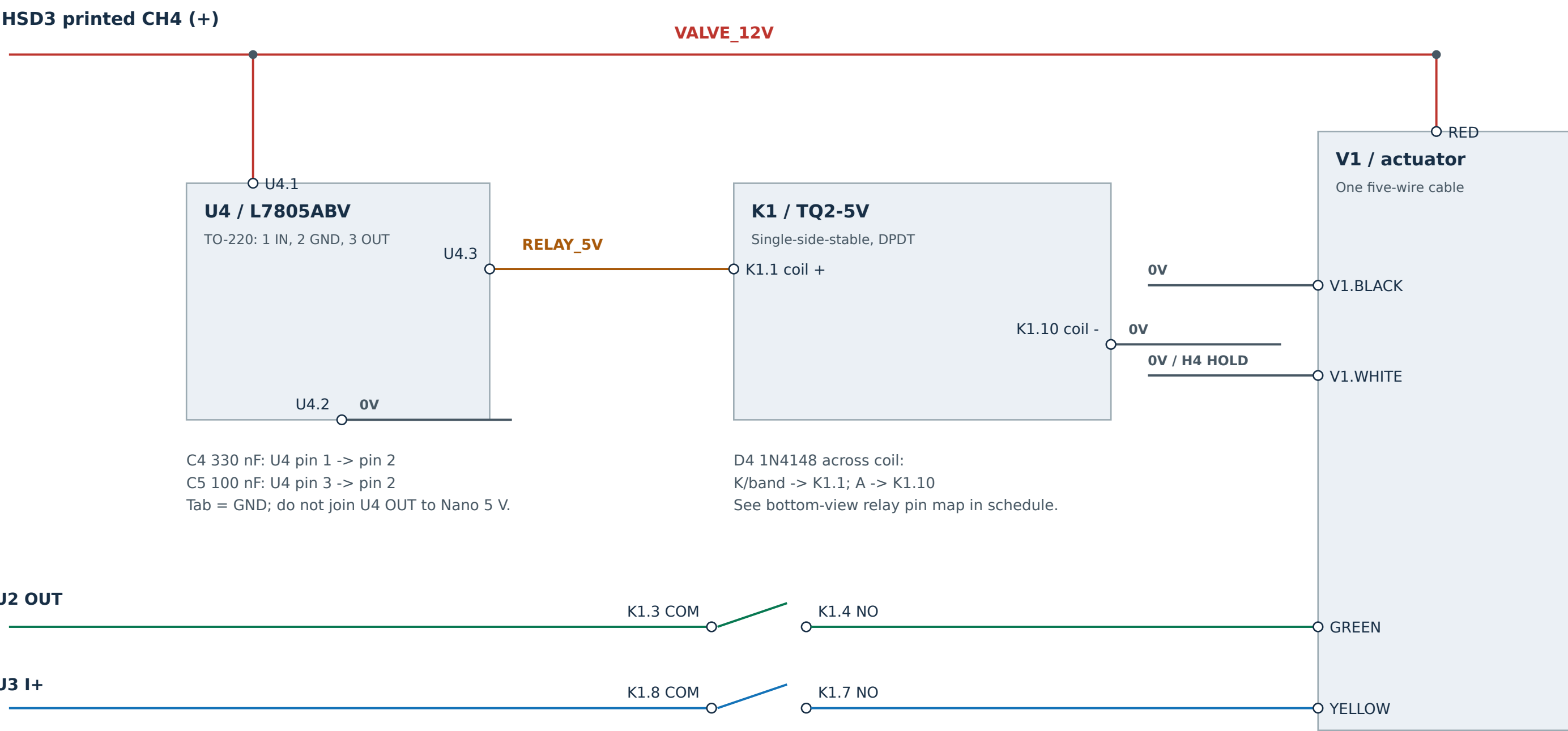
R2, R22 SDA/SCL pull-up jumpers OPEN. Address: HSD1 all open; HSD2 A1 closed; HSD3 A2 closed.
V2 component-side: printed ch7 at top, ch0 at bottom; schematic OUT1 = ch7, OUT8 = ch0.
Top supply: VIN left / G right, Bottom supply: G left / VIN right. Verify markings before power.

BENCH DESIGN — RELEASE HOLDS H1-H5

Functional terminal labels; X1 numbers are NOT machine connector cavities.

03 / VALVE POWER AND SIGNAL DISCONNECT

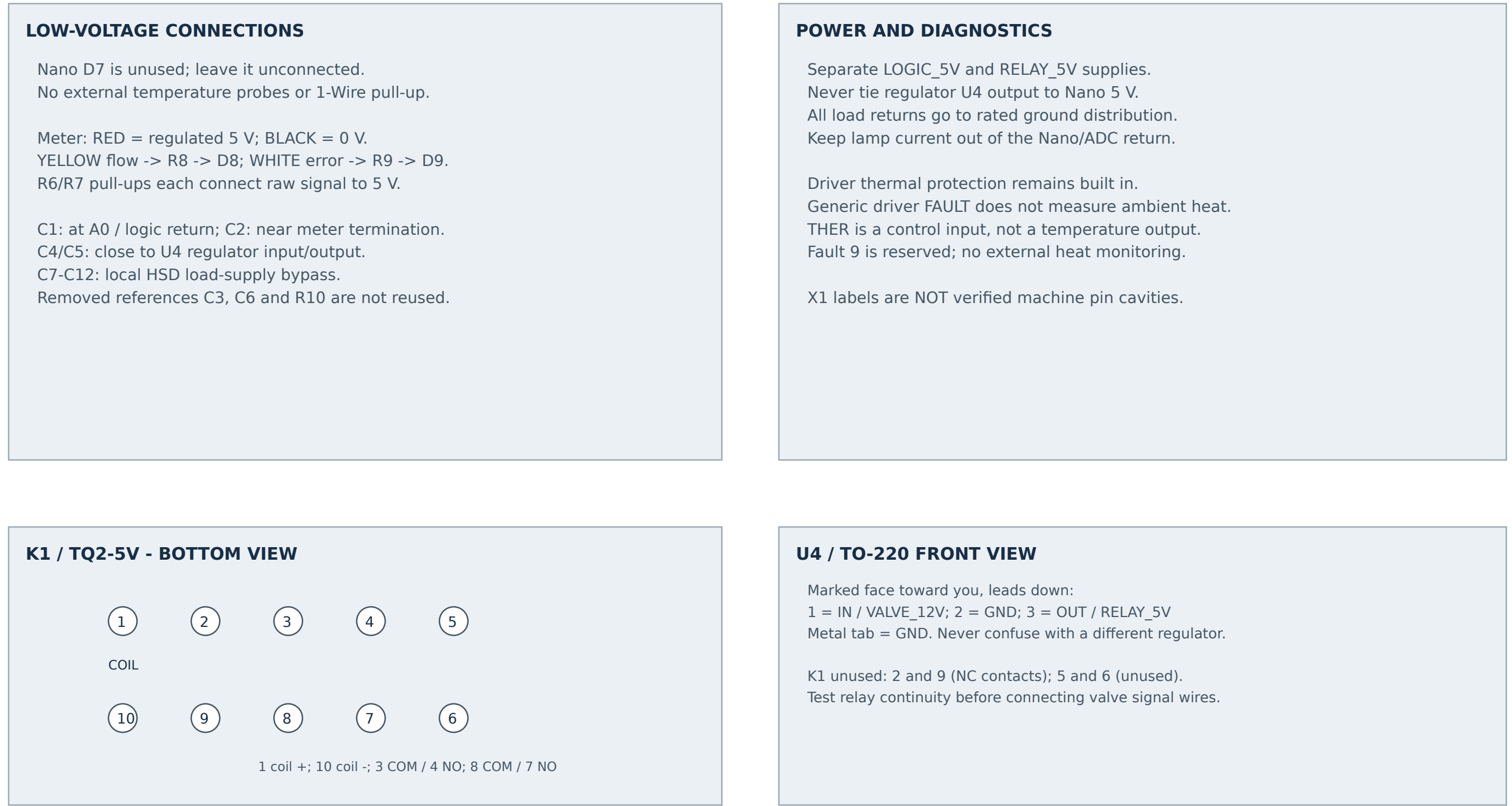
K1 normally-open contacts prevent command/feedback backfeed when the actuator is unpowered



K1 coil de-energized: 3-4 OPEN and 8-7 OPEN. K1 pins 2, 5, 6, 9 are unused.
U2 OUT_GND and U3 I- -> valve white/common -> 0V (return compatibility requires H4).
Proposed watchdog: 500 ms timeout; latch LOW until deliberate closed-command recovery.
Removing power can leave the ball open. Upstream manual water shutoff remains necessary.

06 / TERMINAL ORIENTATION AND ASSEMBLY NOTES

Physical pin references apply ONLY to the stated package and revision



EVERY EXTERNAL CONNECTION / 126 CONDUCTORS										Matching net labels join electrically across sections. Dot = junction; crossing without dot = no connection.										
WIRE	NET	FROM	TO	WIRE	NET	FROM	TO	WIRE	NET	FROM	TO	WIRE	NET	FROM	TO	WIRE	NET	FROM	TO	
W001	MACHINE_12V	TB12.rail	X1.1	W033	SDA	U1.A4	HSD1.D	W065	L4_BLUE	HSD1.CH6	L4.BLUE	W097	VALVE_FEEDBACK	V1.YELLOW	K1.7	W002	MACHINE_12V	TB12.rail	U1.VIN	
W002	MACHINE_12V	TB12.rail	U1.VIN	W034	SDA	U1.A4	HSD2.D	W066	L4_WHITE	HSD1.CH7	L4.WHITE	W098	FEEDBACK_IN	U3.I+	U3.I+	W003	MACHINE_12V	TB12.rail	HSD1.VIN	
W003	MACHINE_12V	TB12.rail	HSD1.VIN	W035	SDA	U1.A4	HSD3.D	W067	0V	TB0.rail	L4.BLACK	W099	POSITION_RAW	U3.SIGNAL	R4.1	W004	MACHINE_12V	TB12.rail	HSD2.VIN	
W004	MACHINE_12V	TB12.rail	HSD2.VIN	W036	SCL	U1.A5	U2.SCL	W068	L5_BLUE	HSD2.CH0	L5.BLUE	W100	POSITION_ADC	U1.A0	R4.2	W005	MACHINE_12V	TB12.rail	HSD3.VIN	
W005	MACHINE_12V	TB12.rail	HSD3.VIN	W037	SCL	U1.A5	HSD1.C	W069	L5_WHITE	HSD2.CH1	L5.WHITE	W101	POSITION_ADC	U1.A0	R5.1	W006	0V	TB0.rail	U1.GND1	
W006	0V	TB0.rail	U1.GND1	W038	SCL	U1.A5	HSD2.C	W070	0V	TB0.rail	L5.BLACK	W102	POSITION_ADC	U1.A0	C1.1	W007	0V	TB0.rail	U2.GND	
W007	0V	TB0.rail	U2.GND	W039	SCL	U1.A5	HSD3.C	W071	L6_BLUE	HSD2.CH2	L6.BLUE	W103	0V	U1.GND1	R5.2	W008	0V	TB0.rail	U3.GND	
W008	0V	TB0.rail	U3.GND	W040	MACHINE_G	R1.1	R1.1	W072	L6_WHITE	HSD2.CH3	L6.WHITE	W104	0V	U1.GND1	C1.2	W009	0V	TB0.rail	U4.GND	
W009	0V	TB0.rail	U4.GND	W041	OPTO_G	R1.2	O1.IN1	W073	0V	TB0.rail	L6.BLACK	W105	LOGIC_5V	FM1.RED	R6.1	W010	0V	TB0.rail	HSD1.LOAD_GND	
W010	0V	TB0.rail	HSD1.LOAD_GND	W042	OPTO_G	R1.2	O1.K	W074	L7_BLUE	HSD2.CH4	L7.BLUE	W106	FLOW_RAW	FM1.YELLOW	R6.2	W011	0V	TB0.rail	HSD2.LOAD_GND	
W011	0V	TB0.rail	HSD2.LOAD_GND	W043	0V	O1.INPUT_GND	D1.A	W075	L7_WHITE	HSD2.CH5	L7.WHITE	W107	FLOW_RAW	FM1.YELLOW	R8.1	W012	0V	TB0.rail	HSD3.LOAD_GND	
W012	0V	TB0.rail	HSD3.LOAD_GND	W044	G_LOGIC	U1.OUT1	U1.D2	W076	0V	TB0.rail	L7.BLACK	W108	FLOW_LOGIC	FM1.RED	U1.D8	W013	0V	TB0.rail	O1.INPUT_GND	
W013	0V	TB0.rail	O1.INPUT_GND	W045	MACHINE_H	X1.4	R2.1	W077	L8_BLUE	HSD2.CH6	L8.BLUE	W109	LOGIC_5V	FM1.RED	R7.1	W014	0V	TB0.rail	O1.HV_GND	
W014	0V	TB0.rail	O1.HV_GND	W046	OPTO_H	R2.2	O1.IN2	W078	L8_WHITE	HSD2.CH7	L8.WHITE	W110	FLOW_ERROR_RAW	FM1.WHITE	R7.2	W015	0V	TB0.rail	U2.HV_GND	
W015	0V	TB0.rail	U2.HV_GND	W047	OPTO_H	R2.2	D2.K	W079	0V	TB0.rail	L8.BLACK	W111	FLOW_ERROR_RAW	FM1.WHITE	R9.1	W016	0V	TB0.rail	V1.BLACK	
W016	0V	TB0.rail	V1.BLACK	W048	0V	O1.INPUT_GND	D2.A	W080	L9_BLUE	HSD3.CH0	L9.BLUE	W112	FLOW_ERROR_LOGIC	FM1.RED	U1.D9	W017	0V	TB0.rail	V1.WHITE	
W017	0V	TB0.rail	V1.WHITE	W049	H_LOGIC	O1.OUT2	U1.D3	W081	L9_WHITE	HSD3.CH1	L9.WHITE	W113	LOGIC_5V	FM1.RED	C2.1	W018	0V	TB0.rail	U3.I-	
W018	0V	TB0.rail	U3.I-	W050	MACHINE_J	X1.5	R3.1	W082	0V	TB0.rail	L9.BLACK	W114	0V	FM1.BLACK	C2.2	W019	0V	TB0.rail	U3.I+	
W019	0V	TB0.rail	U3.I+	W051	OPTO_J	R3.2	O2.IN1	W083	L10_BLUE	HSD3.CH2	L10.BLUE	W115	MACHINE_12V	HSD1.VIN	C7+	W020	0V	TB0.rail	FM1.BLACK	
W020	0V	TB0.rail	FM1.BLACK	W052	OPTO_J	R3.2	D3.K	W084	L10_WHITE	HSD3.CH3	L10.WHITE	W116	MACHINE_12V	HSD1.VIN	C10.1	W021	0V	TB0.rail	K1.10	
W021	0V	TB0.rail	K1.10	W053	0V	O2.INPUT_GND	D3.A	W085	0V	TB0.rail	L10.BLACK	W117	0V	HSD1.LOAD_GND	C7-	W022	0V	TB0.rail	U1.SV	
W022	0V	TB0.rail	U1.SV	W054	J_LOGIC	U2.OUT1	U1.D4	W086	VALVE_12V	HSD3.CH4	V1.RED	W118	0V	HSD1.LOAD_GND	C10.2	W023	LOGIC_5V	TB5.rail	U2.VCC	
W023	LOGIC_5V	TB5.rail	U2.VCC	W055	0V	O2.INPUT_GND	O2.IN2	W087	VALVE_12V	HSD3.CH4	U4.1	W119	MACHINE_12V	HSD2.VIN	C8+	W024	LOGIC_5V	TB5.rail	U3.VCC	
W024	LOGIC_5V	TB5.rail	U3.VCC	W056	L1_BLUE	HSD1.CH0	L1.BLUE	W088	RELAY_5V	U4.3	K1.1	W120	MACHINE_12V	HSD2.LOAD_GND	C11.1	W025	LOGIC_5V	TB5.rail	U3.VCC	
W025	LOGIC_5V	TB5.rail	U3.VCC	W057	L1_WHITE	HSD1.CH1	L1.WHITE	W089	RELAY_5V	U4.3	D4.K	W121	0V	HSD2.LOAD_GND	C8-	W026	LOGIC_5V	TB5.rail	HSD1.V	
W026	LOGIC_5V	TB5.rail	HSD1.V	W058	0V	TB0.rail	L1.BLACK	W090	RELAY_5V	U4.3	C5.1	W122	0V	HSD2.LOAD_GND	C11.2	W027	LOGIC_5V	TB5.rail	HSD2.V	
W027	LOGIC_5V	TB5.rail	HSD2.V	W059	L2_BLUE	HSD1.CH2	L2.BLUE	W091	0V	K1.10	D4.A	W123	MACHINE_12V	HSD3.VIN	C9+	W028	LOGIC_5V	TB5.rail	HSD3.V	
W028	LOGIC_5V	TB5.rail	HSD3.V	W060	L2_WHITE	HSD1.CH3	L2.WHITE	W092	0V	U4.2	C5.2	W124	MACHINE_12V	HSD3.VIN	C12.1	W029	LOGIC_5V	TB5.rail	O1.HV	
W029	LOGIC_5V	TB5.rail	O1.HV	W061	0V	TB0.rail	L2.BLACK	W093	0V	U4.2	C4.2	W125	0V	HSD3.LOAD_GND	C9-	W030	LOGIC_5V	TB5.rail	O2.HV	
W030	LOGIC_5V	TB5.rail	O2.HV	W062	L3_BLUE	HSD1.CH4	L3.BLUE	W094	VALVE_12V	U4.1	C4.1	W126	0V	HSD3.LOAD_GND	C12.2	W031	LOGIC_5V	TB5.rail	FM1.RED	
W031	LOGIC_5V	TB5.rail	FM1.RED	W063	L3_WHITE	HSD1.CH5	L3.WHITE	W095	DAC_OUT	U2.OUT	K1.3						W032	SDA	U1.A4	U2.SDA
W032	SDA	U1.A4	U2.SDA	W064	0V	TB0.rail	L3.BLACK	W096	VALVE_COMMAND	K1.4	V1.GREEN									

COMPONENT SCHEDULE / ALL 49 REFERENCES										Resistors: 1% • ceramic bypass: X7R 50 V • electrolytics: 35 V / 105 °C • diode band identifies cathode K									
X1	Reused machine harness; verify actual cavities	HSD3	Serial Wombat PCB0046 HSD V2	L3	Nilight TL-248BW	D1	1N4148, axial	R8	1000 ohm 1% 0.25 W	C4	330 nF X7R 50 V								
TB12	12 V distribution	O1	SparkFun BOB-09118 v1.2	L4	Nilight TL-248BW	R2	1000 ohm 1% 0.5 W	R9	1000 ohm 1% 0.25 W	C7	47 uF 35 V 105 C electrolytic								
TB5	Logic 5 V distribution	O2	SparkFun BOB-09118 v1.2	L5	Nilight TL-248BW	D2	1N4148, axial	D4	1N4148, axial	C8	47 uF 35 V 105 C electrolytic								
TB0	Ground distribution	V1	U.S. Solid USS-MSV50030	L6	Nilight TL-248BW	R3	1000 ohm 1% 0.5 W	C1	100 nF X7R 50 V	C9	47 uF 35 V 105 C electrolytic								
U1	Arduino Nano Every ABX00028	FM1	SciSense UFM-02-03NP4	L7	Nilight TL-248BW	D3	1N4148, axial	C2	100 nF X7R 50 V										
U2	DFRobot DFR1229	U4	ST L7805ABV TO-220	L8	Nilight TL-248BW	R4	1000 ohm 1% 0.25 W	C5	100 nF X7R 50 V										
U3	DFRobot SEN0262	K1	Panasonic TQ2-5V single-side-stable	L9	Nilight TL-248BW	R5	100000 ohm 1% 0.25 W	C10	100 nF X7R 50 V										
HSD1	Serial Wombat PCB0046 HSD V2	L1	Nilight TL-248BW	L10	Nilight TL-248BW	R6	4700 ohm 1% 0.25 W	C11	100 nF X7R 50 V										
HSD2	Serial Wombat PCB0046 HSD V2	L2	Nilight TL-248BW	R1	1000 ohm 1% 0.5 W	R7	4700 ohm 1% 0.25 W	C12	100 nF X7R 50 V										

INTERNAL COMMONS — no added jumper: U1 GND1 = GND2; U2 GND = OUT_GND; U4 pin 2 = tab; each HSD LOAD_GND = G = G0...G7.

READING: Pin names are functional labels, not footprints. Repeated distribution points mean joined bus terminals. Route lamp/valve returns separately from logic/ADC returns.

Revision D removes external temperature sensing. Use the complete audit for qualification and power-up steps. Power removal does not guarantee water stops.

UNUSED: O1/O2 JP2.1; O2 OUT2; K1 2/5/6/9; HSD3 CH5/6/7; U1 3V3, A1/A2/A3/A6/A7, AREF, RESET1/2, RX/TX, D5/D6/D7/D10/D11/D12/D13.

HOLD POINTS: H1 actual machine cavities; H2 existing fuse / wire ratings; H3 purchased parts; H4 valve signal common / compliance; H5 firmware and bench qualification.